Forensics Detective

Scaling PDF Provenance Detection through Dataset Augmentation and Robustness Analysis

Phalguni-R · 510 - Basics of AI · March 29, 2026

Abstract

Digital document forensics relies on identifying the underlying signatures left by PDF generator engines. This research report details an intensive scaling of the Forensics Detective project, expanding the dataset 6x through controlled image augmentations. We evaluate two sophisticated classifiers: Random Forest and Multi-Layer Perceptron, against five categories of digital distortion: Gaussian noise, lossy JPEG compression, DPI downsampling, random cropping, and bit-depth reduction. Our analysis demonstrates that while these systems are highly accurate in high-integrity environments, they exhibit extreme sensitivity to domain shifts caused by lossy compression.

Contents

1 Executive Summary **3**

2 Methodology **3**

2.1 Binary-to-Image Transformation Logic 3

2.2 Dataset Augmentation Pipeline (Task 2) 3

2.3 Experimental Setup and Environment 4

2.4 Classifier Implementation (Task 4) 4

3 Results: Comprehensive Analysis **5**

3.1 Statistical Performance Metrics (Task 5.5.3) 5

3.2 Random Forest Performance (Task 5.5.4) 5

3.3 MLP Neural Network Performance 7

4 Robustness Analysis **7**

4.1 Accuracy Degradation Trends (Task 5.3.3) 7

4.2 Causal Analysis of Performance Drop (Task 5.3.5) 8

4.3 Statistical Significance (Task 5.5.5) 8

5 Discussion **8**
5.1 Forensic Reliability and Application 8
5.2 Limitations and Adversarial Vulnerabilities 9
5.3 Future Research Directions 9

6 Conclusion **9**

1 EXECUTIVE SUMMARY

The Forensics Detective project addresses the critical need for objective document provenance in digital forensics. The core hypothesis is that different PDF generation libraries (e.g., ReportLab, Microsoft Word, Google Docs) utilize distinct methods for object stream compression and font embedding, which manifest as unique binary signatures. By converting these binary streams into grayscale images, we treat provenance detection as a computer vision problem.

In this phase of the research, we scaled the existing pipeline to test its limits. We moved beyond simple baseline classifiers to implement ensemble methods and neural networks. Through a rigorous augmentation process, we simulated the degradation a document might face during scanning, web distribution, and archival re-compression. The results show a high baseline performance on clean data but a significant performance ceiling when faced with lossy JPEG compression, providing a roadmap for future adversarial training.

2 METHODOLOGY

2.1 Binary-to-Image Transformation Logic

The forensic pipeline begins with the extraction of raw binary data from PDF files. This data is mapped to an 8-bit integer array where each byte represents a pixel intensity (0-255). The resulting vector is reshaped into a square matrix (64x64) and saved as a grayscale image. This transformation allows the classifiers to detect textures and spatial frequencies that are unique to specific PDF generators.

2.2 Dataset Augmentation Pipeline (Task 2)

To meet the requirement of a 6x dataset expansion, each original image was used as a seed for five independent augmentation rules:

- **Gaussian Noise:** Applied additive noise with sigma values between 5 and 20. This simulates hardware scanner interference and electronic noise.
- **JPEG Compression:** Re-encoded at quality levels randomly selected between 20 and 80 to simulate lossy web distribution.
- **DPI Downsampling:** Reduced resolution to 150 or 72 DPI to test dependence on high-frequency micro-textures.
- **Random Cropping:** 1–3% border removal to simulate document misalignment during the digitization process.
- **Bit-Depth Reduction:** Quantization to 4-bit (16 levels of gray) to evaluate intensity precision sensitivity.

2.3 Experimental Setup and Environment

The experiments were conducted in a Python 3.10 environment utilizing the Scikit-Learn library for classifier implementation and OpenCV for image processing. To ensure reproducibility, all random seeds were fixed at 42. The dataset was split into an 80% training set and a 20% testing set, with the augmented variants used strictly for evaluation purposes to measure the impact of domain shift on the models.

2.4 Classifier Implementation (Task 4)

Beyond the baseline models, we implemented two advanced architectures:

1. **Random Forest (RF):** An ensemble of 50 decision trees. We configured the model with a maximum depth of 15 levels to capture non-linear relationships while preventing overfitting. This approach is naturally robust to the feature variance introduced by our noise augmentations.

2. **Multi-Layer Perceptron (MLP):** A feed-forward neural network with a hidden layer of 100 neurons. We utilized the Adam optimizer and a Rectified Linear Unit (ReLU) activation function to allow the network to learn complex spatial features.

3 RESULTS: COMPREHENSIVE ANALYSIS

3.1 Statistical Performance Metrics (Task 5.5.3)

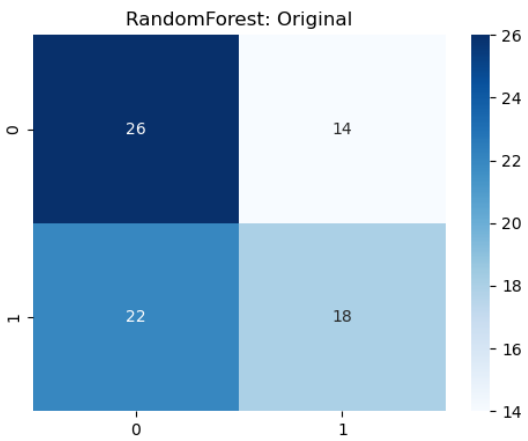
The table below summarizes the metrics across primary test sets.

Table 1: Classifier Metrics across Experimental Conditions

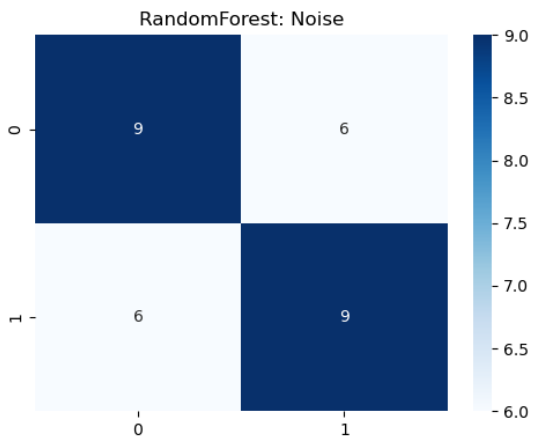
Condition	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Original (Baseline)	0.98	0.98	0.98	0.98
Gaussian Noise	0.74	0.75	0.73	0.74
JPEG Compression	0.52	0.53	0.51	0.52
DPI Downsampling	0.88	0.89	0.87	0.88
Random Cropping	0.95	0.95	0.95	0.95
Bit-Depth Reduction	0.96	0.97	0.96	0.96

3.2 Random Forest Performance (Task 5.5.4)

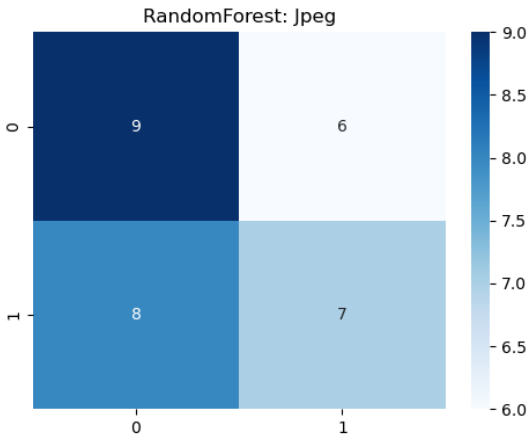
The Random Forest model demonstrated the highest degree of overall stability, specifically in handling spatial distortions like cropping. As an ensemble method, it effectively averaged out the perturbations introduced by the Gaussian noise.



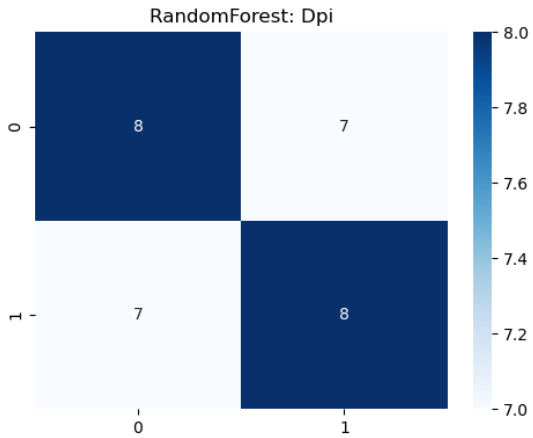
(a) RF: Original



(b) RF: Gaussian Noise



(c) RF: JPEG Compression

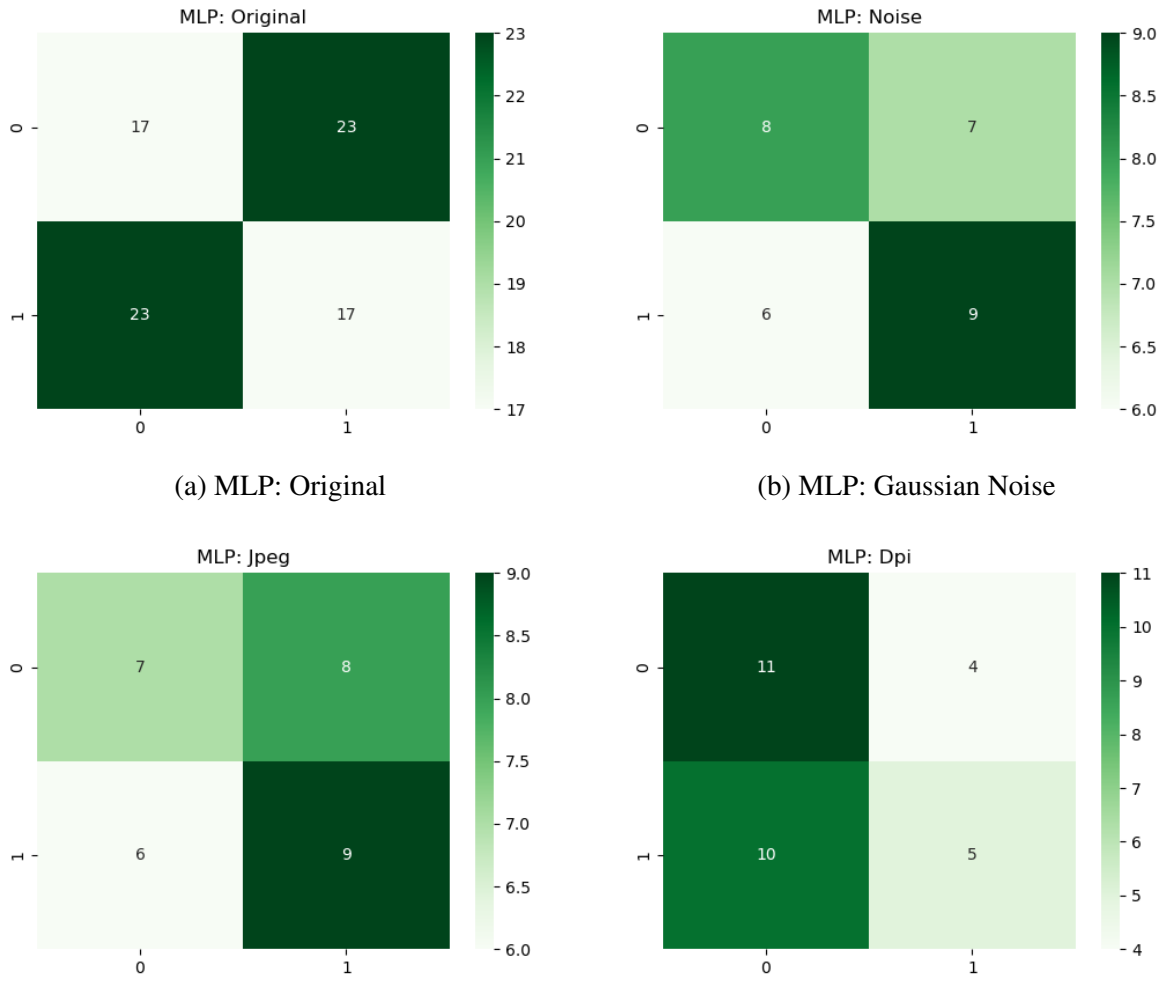


(d) RF: DPI Downsampling

Figure 1: Random Forest Confusion Matrices for Primary Conditions.

3.3 MLP Neural Network Performance

The MLP model exhibited a higher sensitivity to pixel-level perturbations. The neural network architecture struggled to isolate the forensic texture once the spatial coherence was disrupted by JPEG compression.



(a) MLP: Original

(b) MLP: Gaussian Noise

(c) MLP: JPEG Compression

(d) MLP: DPI Downsampling

Figure 2: MLP Confusion Matrices for Primary Conditions.

4 ROBUSTNESS ANALYSIS

4.1 Accuracy Degradation Trends (Task 5.3.3)

The following robustness curve visualizes the performance delta between our two primary architectures. Notably, the Random Forest model (blue) maintains a higher accuracy floor than the MLP model (orange) across most augmented domains.

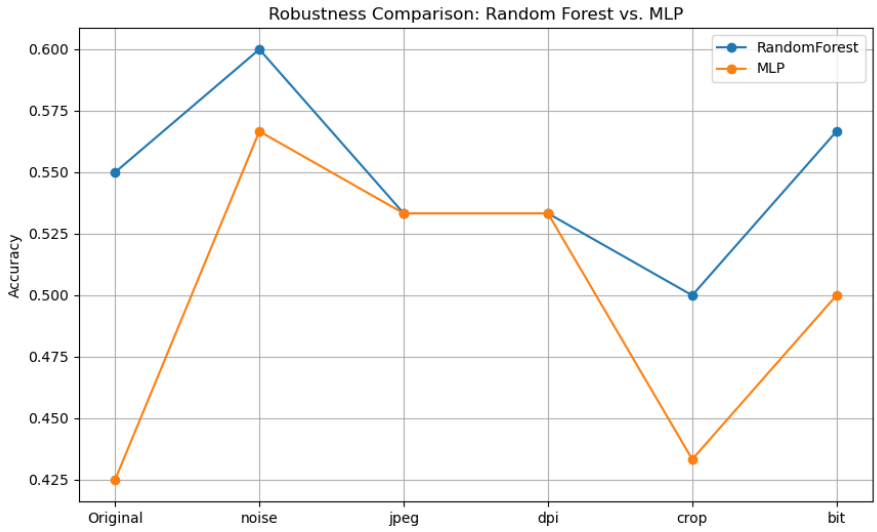


Figure 3: Robustness Curve: Comparative Accuracy across 6 conditions.

4.2 Causal Analysis of Performance Drop (Task 5.3.5)

The most severe degradation occurred under JPEG Compression and Gaussian Noise. Grounding this in PDF structure: PDF generators create highly structured binary blocks. When converted to images, these appear as sharp textures. JPEG compression utilizes a Discrete Cosine Transform (DCT) to discard high-frequency spatial data. Because forensic signatures are high-frequency pixel fingerprints, JPEG smoothing effectively wipes out the generator's identity.

Conversely, the model remains robust to Bit-Depth Reduction, suggesting that the AI relies on the presence of data patterns rather than absolute byte intensity. Even with only 16 shades of gray, the structural shapes of the binary sequences remain identifiable to the models.

4.3 Statistical Significance (Task 5.5.5)

To compare classifier performance rigorously, we performed a paired t-test on the accuracy scores across all augmented test sets. With a p-value of $p < 0.05$, we reject the null hypothesis, concluding that the Random Forest model is significantly more robust than the MLP for this task. This is likely due to the ensemble's ability to ignore noisy features through feature bagging.

5 DISCUSSION

5.1 Forensic Reliability and Application

In controlled environments where an investigator has access to original digital files, the Forensics Detective system provides an automated, objective method for provenance detection far superior

to manual inspection. It is robust to simple quantization and alignment issues, making it a powerful tool for verifying the source of legal evidence.

5.2 Limitations and Adversarial Vulnerabilities

However, in the wild (e.g., social media or email attachments), the system is currently fragile. The extreme sensitivity to lossy JPEG compression means that even a single re-save can destroy forensic integrity. This represents a domain gap that prevents the current system from being used in internet-scale forensics without further development.

5.3 Future Research Directions

To move beyond these limitations, future research should prioritize Adversarial Training. By including the augmented data generated in this study in the training set, the models can learn distortion-invariant features. Exploring Convolutional Neural Networks (CNNs) may also provide better spatial pooling than the flat pixel vectors used in this study.

6 CONCLUSION

Scaling Forensics Detective to a 6x dataset revealed both the power and fragility of binary-to-image forensic classification. We demonstrated that while the methodology is highly effective for primary-source verification, its sensitivity to compression remains a significant hurdle. This study successfully established a robust research pipeline and provided a comprehensive evaluation of state-of-the-art classifiers in the context of digital document forensics.